

WPA2/WPA3 Configuration

The wireless network will use **WPA3-Personal** where supported.

Proposed configuration

Setting	Configuration
SSID	Pentest-Lab
Security	WPA3-Personal
Encryption	AES/CCMP
Authentication	SAE
Password	Strong unique passphrase
WPS	Disabled
SSID Broadcast	Enabled
Guest Network	Disabled unless required

WPA3 provides stronger authentication than older WPA/WEP configurations and protects wireless communication using modern cryptographic mechanisms.

If WPA3 is not supported by the available wireless hardware, **WPA2-AES** will be used as the fallback configuration. An unauthorized device attempting to connect without the correct credentials should fail authentication.

The wireless password will not be stored directly in the project's Git repository.

WIPS Implementation

A **Wireless Intrusion Prevention System (WIPS)** will be used to monitor wireless activity and identify unauthorized devices or wireless access points.

The WIPS will monitor the Pentest-Lab wireless network for events such as:

- Unauthorized wireless clients

- Rogue access points
- Unknown SSIDs
- Unauthorized association attempts
- Suspicious wireless activity

WIPS Prevention Example

The WIPS will be configured to recognize the authorized **Pentest-Lab** network and its approved devices.

For example:

Authorized Device:

Testing Laptop

MAC Address:

AA:BB:CC:DD:EE:FF

SSID:

Pentest-Lab

Status:

AUTHORIZED

If an unknown device attempts to access the wireless network:

Unknown Device

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WIPS Detection

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Unauthorized Client

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Alert Generated

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WIPS Prevention

Depending on the WIPS platform, prevention may include blocking/deauthenticating the unauthorized client, quarantining it through network controls, or alerting the administrator to take action.